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Automation Platform for AESO Interconnection

Group 9 – Final Report

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Original Project Scope

Project

P1828 – Alderson Solar Power Project | 100 MW Solar Facility

Connection: T-tap onto 138 kV Line 100L (Suffield 895S ↔ Tilley 498S)

via new Alderson Park 1061S substation

Electrical Scope

1. Power Flow Analysis *(Steady-State Load Flow)*
Load flow solved using Full Newton-Raphson method
Perform N-1 contingency analysis
Outputs: Bus Voltage Violations, Equipment Overloading Percentages

2. Short Circuit Analysis
Simulate 3-phase & Line-to-Ground (LG) faults using IEC 60909 standard
Outputs: Fault current levels at substations near the point of connection

3. Transient Stability Analysis
Load AESO dynamic models into PSS/E
Simulate disturbance: Bus Fault (3-phase)
Generate output plots: Rotor Angles, Active/Reactive Power, Bus Voltages

4. PV Voltage Stability Analysis
Generate PV curves to identify maximum loadability
Detect voltage collapse point of the system under study
Outputs: PV curve plots exported via matplotlib

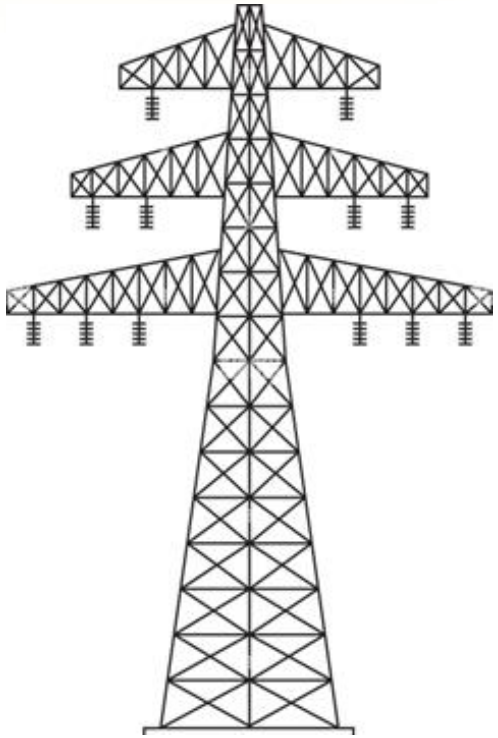
Software Scope

1. PSS/E (Power Systems Simulator for Engineers)
Siemens PTI — industry-standard transmission planning software
Versions: PSS/E v35
Used as the core simulation engine for all 4 study types
Interfaced via native Python API (psspy)

2. Python 3.x — Automation Layer
psspy — PSS/E Python API to control simulations programmatically
pandas — data extraction, manipulation, and Excel report export
matplotlib — output plot generation (used across all 4 modules)
openpyxl — Excel writing engine via pd.ExcelWriter
numpy — numerical computations (Power Flow & Short Circuit modules)

3. GUI — Tkinter
Built with tkinter (tk, ttk, filedialog, messagebox)
threading & queue — run simulations in background without freezing UI
subprocess — auto-open output folders after study completion
logging — real-time log display in the GUI panel

Original Project Deliverables



Must Have

Power Flow Automation Script — steady-state load flow, N-1 contingency analysis, voltage & power data extraction

Short Circuit Study Script — symmetrical & asymmetrical fault simulations, fault current extraction, severity ranking

Transient Stability Script — dynamic simulations, time-domain response extraction, stability assessment plots

PV Voltage Stability Script — PV curve generation, voltage collapse identification per AESO requirements

Data Processing & Analysis Modules — contingency ranking, voltage deviation assessment, thermal loading analysis

Validation Test Cases & Documentation — benchmark testing, comparison with manual PSSE studies, user manual

Should Have

Plotting & Visualization Tools — automated voltage profiles, fault current comparisons, transient stability curves

Result Generation & Reporting Module — summary tables, AESO-compliant Excel/PDF export, annotated plots

Nice to Have

User Interface — simple graphical or menu-driven interface for study selection, parameter input, and real-time progress monitoring

Original Technical Specifications

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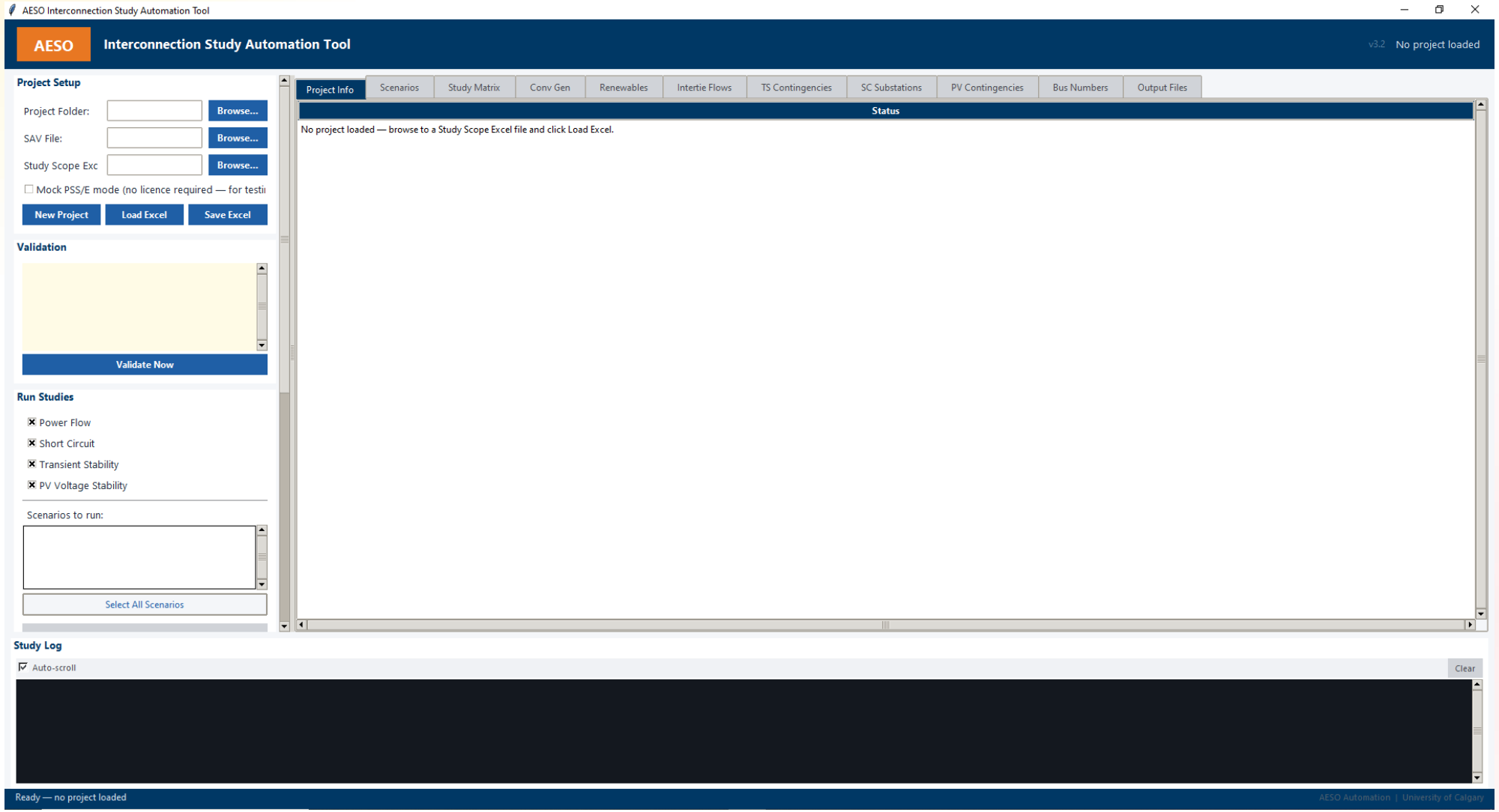
Item	Specification
Software	**PSS/E v35 (Siemens PTI)** — Power Systems Simulation for Engineers. Industry-standard simulation tool for power flow, short-circuit, and dynamic stability analysis of large-scale transmission networks.
Programming Tools	**Python 3.7** with the PSS/E API (psspy) as documented in the PSS/E Application Program Interface (API) Manual v35. **NumPy** — numerical computation and array operations. **Pandas** — data framing, manipulation, and tabular result export. **Matplotlib** — data visualization and plot generation. **openpyxl / xlsxwriter** — automated Excel report writing. **tkinter** — graphical user interface (GUI) for the automation tool.
Computing Hardware	Laptop/Desktop (16 GB RAM, SSD, Intel i5/i7 or equivalent) running Windows 10/11 (required for PSS/E v35 compatibility).
Power Flow Method	Full Newton-Raphson (NR) method via PSS/E FNSL solver. Flat-start initialization applied per scenario. Convergence tolerance: default PSS/E mismatch threshold (0.1 MVA). Category A (N-0) limits: 0.95–1.05 pu; Category B (N-1) post-contingency limits: 0.90–1.10 pu per AESO Interconnection Study Scope.
Short-Circuit Method	Classical symmetrical fault analysis using PSS/E IECS module (IEC 60909 compliant). Four fault types computed per substation bus: three-phase (3PH), single line-to-ground (LG), line-to-line (LL), and double line-to-ground (LLG). Equipment withstand limit: **63 kA** per AESO criteria.
Transient Stability	Time-domain dynamic simulation using PSS/E ESTR/TPLU dynamic engine. Time step: **1/120 s (½ cycle at 60 Hz)**; simulation window: **10 s**; fault applied at **t = 1.0 s**. Per-contingency fault clearing times sourced from the study scope (near-end cycles per AESO Table 2-3). AESO pass criteria: rotor angle < 180° (first-swing), POI voltage recovery ≥ **0.90 pu** within **1.0 s** post-fault clearing.
PV Voltage Stability	PV curve generation via repeated power flow solves with MW injection incremented at the POI bus in **10 MW steps** up to MARP × 1.05. Nose-point (voltage collapse) detected automatically. Category A (N-0) voltage floor: **0.95 pu**; Category B (N-1): **0.90 pu** per AESO criteria.
Automation Framework	Custom Python automation tool with a tkinter GUI (v3.0). Batch execution across multiple scenarios defined in a structured Excel scope file (study_scope_data.xlsx). Results exported automatically to Excel, PDF reports, and PNG plots per study. All AESO compliance thresholds centralized in config/settings.py.

Project Accomplishments (UI)

Front End view



Project Accomplishments (UI)

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After loading profile

Project Info | Scenarios | Study Matrix | Conv Gen | Renewables | Intertie Flows | TS Contingencies | SC Substations | PV Contingencies | Bus Numbers | Output Files

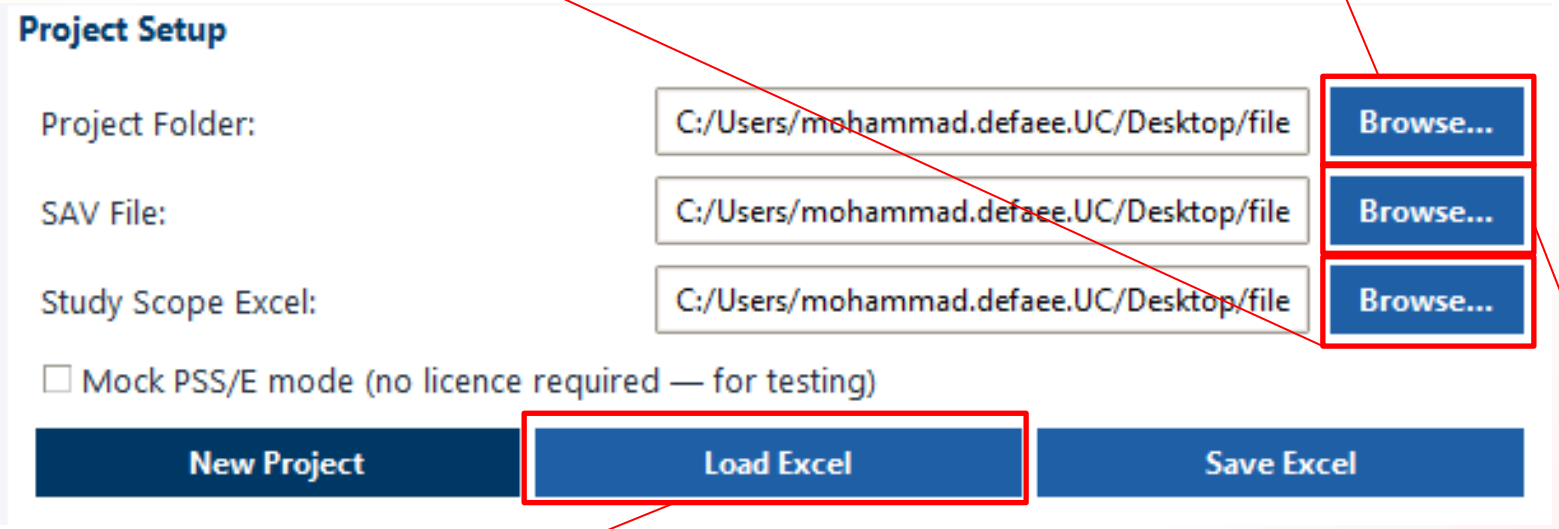
Field	Value
Project Number	P1828
Project Name	HEP Energy Alderson MPC Solar
Market Participant	hep Canada SPV 1 Ltd.
Studies Consultant	Hardline Engineering Ltd.
In-Service Date	2024-12-02
Generation Type	Solar
MARP (MW)	100.0
Max Capability (MW)	100.0
Connection Voltage (kV)	138.0
POC Substation	Suffield 895S / Tilley 498S (T-tap on 100L)
Study Area Regions	48, 47, 4, 52
SAV File Path	
Source Bus Number	275
POI Bus Number	275
TS Fault Bus Number	275

Project Accomplishments (UI)

Project Setup

Selecting the Study Folder

Choosing the .Sav file



Selecting the Study Scope

Load into Software

Project Accomplishments (UI)

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Automated Study on All Buses

Run Studies

- [x] Power Flow
- [x] Short Circuit
- [x] Transient Stability
- [x] PV Voltage Stability

Scenarios to run:

Scenario Name
2025 SP Pre-Project-HG
2025 SL Pre-Project-HG
2025 SP Post-Project-HG
2025 SL Post-Project-HG

Select All Scenarios

▶ RUN STUDIES

Project Accomplishments (UI)

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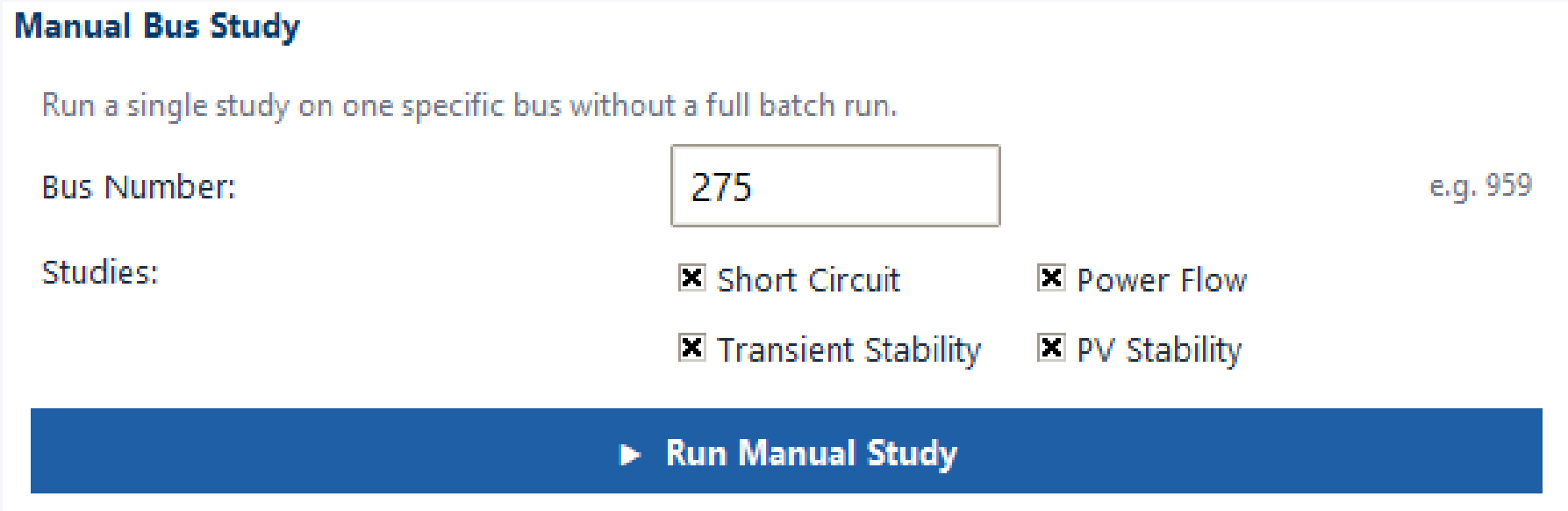
Study on specific Bus



Project Accomplishments (UI)

Outputs

Generates results in MS Exel

Generates Plots for profiles

results/

- power_flow_2025 SP Post-Project-HG_20260412_142919.xlsx
- power_flow_2025 SP Post-Project-HG_20260412_143113.xlsx
- power_flow_2025 SP Post-Project-HG_20260412_143311.xlsx
- power_flow_2025 SP Pre-Project-HG_20260412_142822.xlsx
- power_flow_2025 SP Pre-Project-HG_20260412_143016.xlsx
- power_flow_2025 SP Pre-Project-HG_20260412_143210.xlsx

File / Folder	Size	
results/		
plots/		
power_flow_2025 SP Post-Project-HG_20260412_142919_n1_contingency.png	91.9 KB	2026-04-12 14:29
power_flow_2025 SP Post-Project-HG_20260412_142919_thermal_loading.png	124.3 KB	2026-04-12 14:29
power_flow_2025 SP Post-Project-HG_20260412_142919_voltage_profile.png	3.0 MB	2026-04-12 14:29
power_flow_2025 SP Post-Project-HG_20260412_143113_n1_contingency.png	91.9 KB	2026-04-12 14:31
power_flow_2025 SP Post-Project-HG_20260412_143113_thermal_loading.png	124.3 KB	2026-04-12 14:31
power_flow_2025 SP Post-Project-HG_20260412_143113_voltage_profile.png	3.0 MB	2026-04-12 14:31
power_flow_2025 SP Post-Project-HG_20260412_143311_n1_contingency.png	91.9 KB	2026-04-12 14:33
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power_flow_2025 SP Post-Project-HG_20260412_143311_voltage_profile.png	3.0 MB	2026-04-12 14:33
power_flow_2025 SP Pre-Project-HG_20260412_142822_n1_contingency.png	91.1 KB	2026-04-12 14:28
power_flow_2025 SP Pre-Project-HG_20260412_142822_thermal_loading.png	123.5 KB	2026-04-12 14:28
power_flow_2025 SP Pre-Project-HG_20260412_142822_voltage_profile.png	3.0 MB	2026-04-12 14:28
power_flow_2025 SP Pre-Project-HG_20260412_143016_n1_contingency.png	91.1 KB	2026-04-12 14:30
power_flow_2025 SP Pre-Project-HG_20260412_143016_thermal_loading.png	123.5 KB	2026-04-12 14:30
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power_flow_2025 SP Pre-Project-HG_20260412_143210_thermal_loading.png	123.5 KB	2026-04-12 14:32
power_flow_2025 SP Pre-Project-HG_20260412_143210_voltage_profile.png	3.0 MB	2026-04-12 14:32
	333.9 KB	2026-04-12 14:32

File / Folder
results/
plots/
reports/
power_flow_2025 SP Post-Project-HG_20260412_142919.pdf
power_flow_2025 SP Post-Project-HG_20260412_143113.pdf
power_flow_2025 SP Post-Project-HG_20260412_143311.pdf
power_flow_2025 SP Pre-Project-HG_20260412_142822.pdf
power_flow_2025 SP Pre-Project-HG_20260412_143016.pdf
power_flow_2025 SP Pre-Project-HG_20260412_143210.pdf

Give reports as Pdf

Project Accomplishments (Power Flow)

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Auto VS Manual

Results:

POWER FLOW | Bus 275

Converged: True

V (pu)	Angle (°)	V_min	V_max	Result
1.0217	2.59	0.950	1.050	PASS

Output Bar

```
AESO 22021RC_2025SP_SOUTH-HIGH_TIE-ECON_SOLAR-0.95_P                    RATING   %MVA FOR TRANSFORMERS
***SUBJECT TO CEII NDA*** SEE LONG TITLE FOR DISCLAIMER                 SET  1   % I  FOR NON-TRANSFORMER BRANCHES

BUS    275     TILLEY 7    138.00 CKT     MW     MVAR     MVA   % 1.0217PU    2.59  X--- LOSSES ---X X---- AREA -----X X---- ZONE -----X    275
                                                                   140.99KV              MW     MVAR   47 BROOKS         99 NONCLASS
 TO SWITCHED SHUNT                        0.0    -0.0     0.0
 TO    270     SUFFIEL7    138.00  00   -34.7   -22.2    41.2  52                      0.79    1.83    4 MEDICINE       99 NONCLASS
 TO    276     W BROOK7    138.00  66    13.2    12.4    18.1  15                      0.12    0.27   47 BROOKS         99 NONCLASS
 TO   3275     TILLEY 8    25.000  T2    13.3     5.1    14.2  34 0.9552RG             0.02    0.67   47 BROOKS         99 NONCLASS
 TO   4275     TILLEY 9    25.000  T1     8.2     4.7     9.5  23 0.9670RG             0.01    0.30   47 BROOKS         99 NONCLASS
```

Project Accomplishments (Short Circuit)

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Auto VS Manual

Results:

```
SHORT CIRCUIT  |  Bus 275

Fault   I_fault (kA)  Limit (kA)  Result
------  ------------  ----------  ------
3PH            0.000        63.0  PASS
LG             0.000        63.0  PASS
LL             0.000        63.0  PASS
LLG            0.000        63.0  PASS

[Info] Running full-system power flow to extract Bus 275 result…
```

Output Bar

```
THREE PHASE FAULT AT BUS    275     [TILLEY 7    138.00]:

SEQUENCE              RE(V0)    IM(V0)    RE(V+)    IM(V+)    RE(V-)    IM(V-)   RE(3V0)   IM(3V0)
PHASE                 RE(VA)    IM(VA)    RE(VB)    IM(VB)    RE(VC)    IM(VC)

   275     (P.U.)     0.0000   -0.0000    0.0000    0.0000   -0.0000    0.0000    0.0000   -0.0000
TILLEY 7    138.00   0.0000   -0.0000   -0.0000   -0.0000   -0.0000    0.0000
```

Project Accomplishments (Transient Stability)

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Results:

TRANSIENT STABILITY | Bus 275

Contingency	Angle	V_poi	Result
Contingency Name	0.0	1.0000	PASS
666L (West Brooks 28S - Tilley 498S)	0.0	1.0000	PASS
666L (West Brooks 28S - Tilley 498S)	0.0	1.0000	PASS
892L (Suffield 895S - Bowmanton 244S	0.0	1.0000	PASS
892L (Suffield 895S - Bowmanton 244S	0.0	1.0000	PASS
763L (West Brooks 28S - Vauxhall 158	0.0	1.0000	PASS
763L (West Brooks 28S - Vauxhall 158	0.0	1.0000	PASS
853L (West Brooks 28S - Queenstown 5	0.0	1.0000	PASS

Parameter	Value
Scenario	Scenario Name
SAV File	2021RC_2025SP_South-High_Tie-Econ_Solar-0.95_p.sav
Simulation Duration (s)	10
Contingencies Run	27
AESO PASS	27
AESO FAIL	0
Rotor Angle Limit (deg)	180
Voltage Recovery Limit (pu)	0.9
Recovery Window (s)	1
Reference Bus	101
Reference Gen ID	1

Project Accomplishments (PV Voltage Stability)

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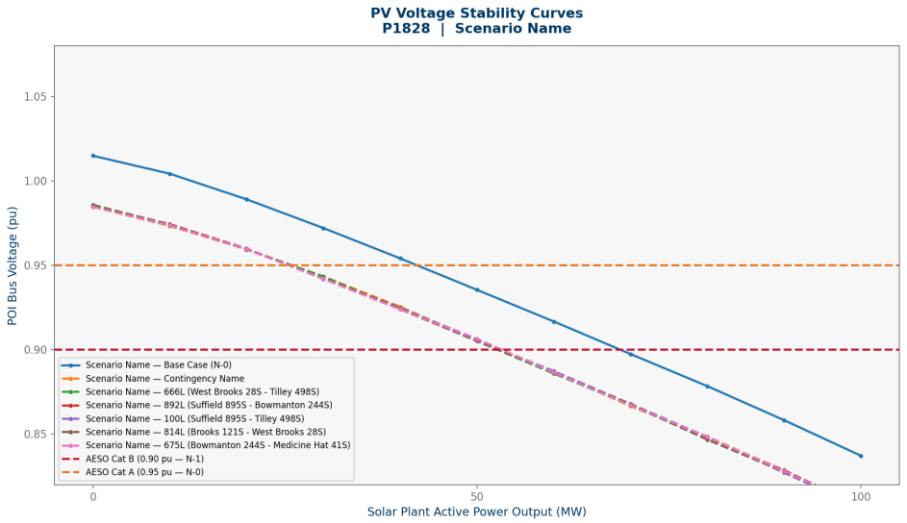
Total Output Summary and Results of Buss 275

Parameter	Value
Scenario Label	Scenario Name
Source (Generator) Bus	
POI (Monitor) Bus	
Transfer Range (MW)	0.0 – 105.0
Step Size (MW)	10
Curves Run	7
AESO Cat A V_min (pu)	0.95
AESO Cat B V_min (pu)	0.9
Most Critical Scenario	Scenario Name — Base Case (N-0)
Critical Collapse MW	Not Reached
Critical Max Stable MW	100
MARP (MW)	100



Results:

Contingency	Collapse	Min V	Result
Manual Bus 275 – Base Case (N-0)	–	0.8372	VIOLATION
Manual Bus 275 – Contingency Name	–	0.8075	VIOLATION
Manual Bus 275 – 666L (West Brooks	–	0.8084	VIOLATION
Manual Bus 275 – 892L (Suffield 895	–	0.8069	VIOLATION
Manual Bus 275 – 100L (Suffield 895	–	0.8075	VIOLATION
Manual Bus 275 – 814L (Brooks 121S	–	0.8073	VIOLATION
Manual Bus 275 – 675L (Bowmanton 24	–	0.8080	VIOLATION

Validation and Measures of Success

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Measures of Success:

- Team Proficiency in PSSE: All team members proficient, verified through practical application.

Validation Tests Results:

- Team Knowledge Assessments: Extensive case studies and simulations to ensure team capability.
- PSSE Script Functionality Trials: Multiple trial runs confirm automation script reliability.

Code Logic Validation:

- Peer Reviews: Ensured diagram clarity and correctness.
- Simulation Tests: Verified script performance under various conditions.

Python Script Development Success:

- Robust Script Execution: Power flow, short circuit, transient stability.
- Efficient Data Handling: Automated extraction from PSSE outputs.
- Scalable for Large Studies: Proven capability in large-scale applications.

Deviations from Original Goals

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Priority	Deliverable	Sub-requirements	Status	Notes
Must Have	Power Flow Automation	Steady-state load flow, N-1 contingency, voltage & power extraction	✅ **Complete**	Fully implemented in studies/power_flow/; N-1 loop, bus voltage/branch MW extraction, Excel export confirmed
Must Have	Short Circuit Study	Symmetrical & asymmetrical faults, fault current extraction, severity ranking	✅ **Complete**	studies/short_circuit/ covers 3PH, SLG, LL, LLG; severity ranking DataFrame confirmed
Must Have	Transient Stability	Dynamic simulations, time-domain extraction, stability plots	⚠️ **Partial**	Simulation pipeline complete and PDF output generated (as shown in attached report); **Min POI voltage always returns 1.0** due to a bug. The .dyr file doesn't work correctly,
Must Have	PV Voltage Stability	PV curve generation, voltage collapse identification per AESO	⚠️ **Partial**	studies/pv_voltage/ implements incremental load stepping, nose-point detection, N-1 PV curves, couldn't have .ecd file to validate.
Must Have	Data Processing & Analysis	Contingency ranking, voltage deviation, thermal loading	✅ **Complete**	utils/ and core/ handle ranking, deviation scoring, and thermal loading percentage per branch
Must Have	Validation & Documentation	Benchmark testing, comparison with manual PSS/E, user manual	⚠️ **Partial**	Since the base case was not converging for some studies, we validated it with a test case
Should Have	Plotting & Visualization	Voltage profiles, fault current comparisons, TS curves	✅ **Complete**	Multi-panel PNG plots per contingency, voltage profile overlays, fault current bar charts all implemented
Should Have	Result Generation & Reporting	Summary tables, AESO-compliant Excel/PDF export, annotated plots	✅ **Complete**	reporting/ module + save_results() in every study class produces Excel + PDF with AESO-formatted compliance summary tables
Nice to Have	User Interface	Study selection, parameter input, real-time progress monitoring	✅ **Exceeded**	GUI in gui/ + main_window.py (68K chars) with study launcher, progress bar, log window, and parameter panels — exceeds the "simple menu-driven" baseline

DEMO